

Topics in CS: Problem Set 5

Due date: December 11, 2025.

Question 1. (25 points) Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism.

We will prove the first isomorphism theorem, stating that $G/\ker(\varphi) \cong \text{im}(\varphi)$.

Denote $K = \ker(\varphi)$ and the quotient group $G/K = \{gK \mid g \in G\}$.

1. Let $g_1, g_2 \in G$. Prove that if $g_1^{-1}g_2 \in K$ then $\varphi(g_1) = \varphi(g_2)$.
2. Consider the map $f : G/K \rightarrow \text{im}(\varphi)$, where for every $gK \in G/K$ we define $f(gK) = \varphi(g)$. Prove that f is well defined; i.e., that $f(gK)$ does not depend on the representative g .
3. Prove that f is a homomorphism.
4. Prove that f is a 1-1 and onto.

Question 2. (25 points)

1. Let G be a group, let $H \leq G$ be a subgroup. Prove that $H \trianglelefteq G$ if and only if for every $g \in G$ it holds that $gHg^{-1} = H$, where $gHg^{-1} = \{ghg^{-1} \mid h \in H\}$.
2. Consider the sets $G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, c \in \mathbb{R}^*, b \in \mathbb{R} \right\}$ and $H = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in \mathbb{R} \right\}$.
Prove that $G \leq GL_2(\mathbb{R})$ and that $H \trianglelefteq G$.

Question 3. (10 points) Let G be a group and suppose that $|G| = 2n$ for some $n > 0$.

Prove that there exists $g \in G$ such that $\text{ord}(g) = 2$.

Question 4. (10 points) We saw that the subgroup $\langle (123) \rangle \leq S_3$ has index 2 and is a normal subgroup. As promised in class, you will now prove a more general claim.

Let G be a group and let $H \leq G$ be a subgroup. Suppose that $[G : H] = 2$. Prove that $H \trianglelefteq G$.

Question 5. (30 points) So far we have seen several groups of order 6: some are cyclic, e.g., $(\mathbb{Z}_6, +)$ and $(\mathbb{Z}_7 \setminus \{0\}, \cdot)$, and some are not, e.g., S_3 . In this question you will prove that every group of order 6 is isomorphic either to the cyclic group of order 6 or to S_3 .

Let G be a group of order 6.

1. Prove that G has a cyclic subgroup of order 3.
Denote this subgroup by H and let h be a generator of H .
2. Prove that G has an element of order 2. Denote this element by g .
3. Prove that $G = H \cup gH$. Conclude that $g^{-1}hg = h$ or that $g^{-1}hg = h^2$.
4. Prove that if $g^{-1}hg = h$ then G is cyclic.
5. Prove that if $g^{-1}hg = h^2$ then G is isomorphic to S_3 .

Good luck!